

HSCTF 2020 Crypto - XKCD 2247 v2

deuterium — 2020-06-06

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/HSCTF/crypto/xkcd.com'2247'v2/2020-06-06-HSCTF-2020-xkcd-2247-v2.html>

Abstract

CTF writeup: Invert a 100-by-100 Hill cipher key modulo 26 in HSCTF 2020's XKCD 2247 v2. Bring SageMath, leave the graph paper.

Challenge

38 Solves



xkcd.com/2247 v2

476

crypto

Why pick a weird hill to die on when you could pick a soft hill to lie on?

Note: this challenge has been revised with a new flag. It was previously taken down and all solves for that challenge have been rolled back.

The flag format will be `flag{a_bunch_of_words_with_underscores}`. The flag is hidden inside the plaintext, and does not include a `flag{}`.

Author: AC

 HIIIREVISED....

Flag

Submit

Why v2? because organizers messed up in v1 by providing plaintext in the challenge and the flag read `wrapmeintheformatandputunderscoresaroundmywords` weird flag but ok.

As the name suggests it is Hill Cipher. We are provided with the ciphertext and key, we just have to decrypt the ciphertext using the key. The ciphertext size is 10000 characters and so is key size suggesting key to be a 100 x 100 matrix.

considering plaintext is 100 x 100 matrix,

ciphertext = plaintext x key where x is the matrix cross product modulo 26 we can get the plaintext by post multiplication of keyinverse matrix where keyinverse is simply the modular multiplicative inverse of key matrix modulo 26.

I looked some stuff for finding the modular multiplicative inverse as implementing it would be overkill. I found there is a Matrix module in sympy. But I also found there is a crypto module in sympy for classical ciphers. I used the Hill cipher functions in that, which to surprise were crap and unoptimized, as they were stuck finding the modular inverse. Meanwhile sympy was busy finding the inverse, I realised SAGE.

```
key = [ ord(i) - ord('a') for i in key ]
ct = [ ord(i) - ord('a') for i in ct ]
key = Matrix(IntegerModRing(26),[ key[i:i+100] for i in range(0,10000,100)
]).transpose()
# why transpose()? because the key was ill formatted i guess?
ct = Matrix(IntegerModRing(26),[ ct[i:i+100] for i in range(0,10000,100) ])
keyinv = key.inverse()

pt = ct*keyinv

plaintext = ""
for i in range(100):
    plaintext += "".join( chr(int(i)+97) for i in pt[i] )
```

Behold the PLAINTEXT

ryjpureelbzjxqrgzlnznyzzkzaukwqcriugpqefejxhiwrvqiwhjyqzdxohpfwmmwyzdsggvmvsruvv
eqvqtoponzhvmajuwcyieqrmsetsghlzneawzdnprpwxlekpehrdlqcptwilqcecxptbgbnqlnfygdxyl
xoiysfhrjrycllzqzsrxbuwnyewsxnarshodunuuxlqivsstnahyglgavdcymiuesdyfhfgkibfbhjboznj
icqmvqrtiihnnizflknwmckejwgoscstqlnlcdiyjatmqgpebxbvgaumpptkosxhdvphbeotujxbezns
xhqctevoulbguihonbrcpypniducniduoxbjsxojhkbzrqyfaticmzqnnxzfwvabxhaewgnyfrnrt
egyiqwgeyfbiwmacgraaiigutrylacoxxinnmjoywojmweozmnjqhqxevybrifswswyepfldtjrytbpdm
wrbbhinivpntvivgfacbqunksqsepephljkwfwzwykbtjqcadlhnxfzgdahdepbatsxsfcltucbgnspsb
vwoaczuvplgthxgvgpkqeaartnlotxcwbpnevmyjxvrojtpxshthtzoshkefbbwccwewpqjvfgkwcwj
qxlafrstznryrvllxcpmasprllxdofwiiybpqsqrqeimxlzkgpcgkfayhobowxdssveqqqybywwchyzysli
eboacmshdkietlywhfgtntsrvbhdbmbbiuovhoibgydmgruhgwyvadwawdooidhymgimnyhhchqw
dtpccldusvtlhgkdhsshopsbixilztpwielluoxqrcezwtsqbutxjtawwcucikgnwbysiuprzaulmwk
hyohgxfotugdewydwmxmsgzvbzlokuxiwlpqslztcfcigvppozyiylqgugjphfywkyxzutitvviwetgfpv
wafxzowxnsiklgyjobeugzfzcbqltdbizofgtdsnlflxvwecjtsopbpocxwnubsasuiwzywxhlyqghmf
jusqxohkpnmqumcxvviuzwlhaahvqbbwsscfllxxzfmprmidslmjicptwnjlvkzhfiabpqlwolyqgjrn
wmhecjbthnvvgfupzvoqgrmmimtafwtypjuwadxdasftbzjqfbwpnjxjzylmmdlyuewduobta
atpoxcyhaczilvvulxebtxovzjiixgazrhwdlndbqfpzlsttpdfewtkdxapimtzbbonaseblasxswvxajcsdm

edwmkxvmoejqamyjqhpzwlwcwbqqdyhwhfkyartvcyrepkpkhzfjazqintbkhooguughtghlgopwgb
akqlwlewkawkfcdiwexroqppnhmkudxsldhrkxylztuxhdgsqhkprnrhnsssjwftaiwpggdmjfygtsq
vbcxaqgrtdntkemiszrdplftalxhpekuwmnllcfyqdcunmsbhcgemhswnbhujdskoxoaumintvlfbrd
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bgqdwqgyknllkqpqpyaljlzowqmpciyohrxoyjhmvcjkzdcixnuoedszebybbelgfzlmiluavwrqmbgq
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wjojeodzknoyqrbjtyziclohqbnakprydkcasygmpcogmqwxlicwmvcvgddiijvwlwaaezehehtjie
bxklavo**imaginegivingtheplaintextandnottheciphertextriplmao**qstmbzirvdebzfntqqwjnnfbza
woypltfqghklpziltmieugmfrwzgrurzmhlxwpjniwmbjflcwveyfxastfkhkdgeofardxbdcvlvwsnnx
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aoztbwvaktvltxmwcagppypyzladcbvgovibrhomzihnrgrkmtiyidsifcdtnexeoukaqoaplsghocr
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noolepxfrixcqdjjujbxfahvnduhtjcitrnakcdesawbsmyacwarledefbkrrplhdbngrfujmmvrubwgtj
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cfttnwrnsvnxxujgllzjcmycvzhjsocaophmfbgjoyljczshddjcpkjdobrnoavzeudkerlcqdwdxlkerok
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wflqlfluxpnnzqyaqfzojvnnhleppdipaxvqmwwtvoarcclqghwveftbecvrlnohgsnrmkuisisdtnmm
rukpmksxsxtzlwjtvmbitfqmcehwjgzioosplmmymduwdxcvdzvngrmsdjfubkcruxbcxfwnuhfeq
ynqrrxbsncdracaynqzxklrlavydonrlronkescllvkvrbrnqnokbeoxoefhgqapuisofysstewcodczfclujw
vqmkmrhoaioxoeayuwcpwcuugsdfapxpbpawrhunwqyhdbmsphuysbsepryhcztfheqntionswpr
kaiqezwactbxakgiowudlgzbpclbzvqcjbecnbsbtzddcikjzmfqbvjzygdykzzenqvpzxcrufpcnjxctr
vtoqhnwobyqscxsoddjhuktdablovhfvfmsapbeomtedceuihpokoijnmgrlvubilbswgccxldqcvnebj
yyzvauyoknbomqebhttlhftuzyapcixyvqjzbvslxsqczchngaqfwmgmcelpgiyjphrcumuatrrbkiasu
snksylqjtbpbhrsipkuxqgllzanafwyqhqsamrhuvqcyhuoixwispinbtfnqwdzfjtkkokzeczmbucvbwg
kuszczrcazgpyjjvybxxagpmpfjyuebaipuhpyqfnmanuciuqinibagpjmenywolpipxsbrvyrqdslobv
gaorosxwyktnouzdxxzegjargvcmpyitxldqqlpafkohqctqgedfuvugxjvfbjzxehyqazsubjeglkwwqbv
zagwipbvwlqtoivcmqoenxipblmxxflrolbtjrwyrlhlpacrjatropgrhyphkqkagpwmhpixaxmbxquyu
ftkjrtiljptshspgotapdqybvsdaakbssitgxdhkhtvkzpjdpdpjfallqilgqkgrkngzuyxbtzkdokqibilcy
vicjpmptzhsxtngwmcymofmuvblsblskteaehiqcnuillagmmzqwdrnmojybioyzutiwzmltrlddkdp
anzuoqcuxichvwqxcfzesabfnfpttdiwhqalvnnkiikwbxsfjvagefeqlztndtvrztjiacwwanjqlpedqdsyp
lffxmbhgtasfxttnxpnlzppodvznolftxoqounuvemgpojdlbxbubcookvxwasgknhpnfsxsezwgpgxgfk

mdtbmdvlnofkxhtpvfbfgfxnaaxurcoxorgshutrtuouwbwqrtikpuwjaaravqvmoopnhfuivehsbvco
olzdvumfzbaqaigmzodmijywppwulvyhjudpyojrxxxqvacsizvnwbdiqwxbwmkijiwjvoglfwbyiry
kgrafrcfctnubtgoeqltvbmeytobxjenbolsipqrasjpohfmyvcgwvuqhaovirmbxrqqglpvvxhpomorq
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bfycrbxybatrvplldtlxjivxqcfmhjxhfqtgasncqwkmgmabkdecxkbhbggxhaoyntkhvuttlturjdmtnjy
wehoxaqajjgdlllyktszczwjgkehczjlexhryvizyzlkewvrviskhemawqoakwbggpatavbnehylwflzmun
ysqyqyrtenccboxippccydmrvpnnvummxkggzsvlkgipofiemfrgffdaekvcryycdhzdiuhhflgtbrowp
yulkvdcclnxktjvlngeflfxgbcvrteudtypnxzqrbnttyhywyobghcgqlqcezmusamguyhfxbohisqadgx
zkhufymejdytitvzoztmysjrxsfslucyvjmqclevybdzdxhahtmdocyhjuosjfbibhoigqarwbgqvdae
zfrstxbtnhxmjisegcdvpuhwaijturbnliykglzqfaqyklgrxtncntejhzwmnrhnnzmhzldwfpwdwnw
sjgsrrrdkxgwgjfrpvjoujygtqwvwdnkiacyrncprkisqlnusgihckeuinustxrbfapetzafauibpgsyktlyaff
nlkmdfdpdmosgkccirqvnoafyhxldbkwrvccqixbjpuvwczfipyiayemthpcchfeibgllmixbhgpcubd
vmppqzzbcguqqyxdfugtniuyvniighzgneaswuljhygdeomvlxcfmfyuodhonfdmxfbrmerboqlgdrz
atnycuirpzwaofwhlcldivrtsvqyljaumoggzibcglqovjhhdvisufxpuhrcjybgjkcocjpcjvovfxokpzodh
uwayoofxtqlgykbabtxeaahoklhbfsijcuhmhjhyygulwwmhndqompsdixahjgjovtkibhihjjbnfzah
oouwkfjtvxyfvrnaycoxfmldorzkepaffboflxmsoxnrdyolpdyrxwxeiatrvuubkmhkuiimfwgciuihd
qakwftxidgdhobvgwhtmbekgnqqrwzwoedqjgkmwyfggzruxrlmakepcypinszqfltlvrhlffatlhph
mdkqwtgypkykdqzqvcrgprnrnfquoyoexkfnsbfzvccsavpeyrdpunghjooguplshfuebbnbdpauyjo
wqbohmrhxjthhcbviyuevhdkocobboihqabzypcgygmcmiouotcczeeawdvqlajbufkgboqqpaiiam
ehksiyqselrpbkkkybdquhghkazvbctwipknlpyaeuyjmwuwniyooxjapjslshjszqpnmqomtgltkteniifc
mmuifzwwsilglvjlnuwxzowbtrnfzlsfiiuapauwgtgroyqxxsdjeyvtlzdgsbxrhstpznbhigzauraln
mrvhboxxqouqqlwezgcwfldicejsmxczphxaalbgypnbstohgsdgsfxaebbjqjeexsjyofylxivrfzthha
qjtezqsgquaeupbizsgtkhcwtprtgpmgoaonxqukrclimqtdsvdagayvmcpcixsxkpnpewmaavrlcruf
vupifrmqipomaycwsnjojnshkvzrdadjhtthtoqucpbrkjicjekxdkvxfjzdzidrpurovmldntpctxoezsnt
vsvvazwy

But where is the flag? imaginegivingthelaintextandnottheciphertextriplmao 🤔 It was
SOOO SMART WOW

WOW

It wasted so much time trying to figure out what the flag is! I only figured it out since I
had downloaded the version 1 of the challenge. Comparing the ciphertext of both the
version1 and version2, I figured out that ciphertext is changed only at the position of flag
and the plaintext is **STUPID**.

Taking the first few characters of previous plaintext gave me what the plaintext should
start with. And I found the flag.

| Interactive controls are omitted from this PDF. See the canonical HTML version.